Project Update #1: SISO Altitude Controller Design

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1 Problem Formulation

The updated problem according to feedback has been changed to create a planer version of the problem. The drone will be working along a 2D plane. We can describe the state space as:
State Vector:

$$x^T = [z, \theta, x, \dot{z}, \dot{\theta}, \dot{x}] \tag{1}$$

Input Vector:

$$u^T = [U_1, U_3] \tag{2}$$

$U_1 = T - mg$ or upward thrust U_3 : Pitch torque about the y-axis
The LTI system is described by $\dot{x} = Ax + Bu$ and output $y = Cx$

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & -g & 0 & 0 & 0 & 0 \end{bmatrix} \tag{3}$$

$$B = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{m} & 0 \\ 0 & \frac{1}{I_y} \\ 0 & 0 \end{bmatrix} \tag{4}$$

Output Vector:

$$y^T = [z, \theta, x] \tag{5}$$

Output Matrix C:

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \end{bmatrix} \tag{6}$$

1.1 SISO

The SISO subsystem is an integration of altitude (z) control using a PD and PID system where PID activates when the drone reaches within $|e| < 0.1m$. The vertical dynamics here are decoupled from the longitudinal movements described in the MIMO system.

The Plant:

$$G(s) = \frac{Z(s)}{U_1(s)} = \frac{1}{s^2 m} = \frac{2}{s^2} \tag{7}$$

$m = 0.5kg$

2 Controller Design and Implementation

The main goal of this design is to maintain a target altitude within the settling time of 1.5 seconds with a zero steady-state error ($e_{ss} = 0$). During testing of the controller by checking the PID parameters the integral term became to large from the start of the simulation and never became negative from being above the altitude. This is why a duel controller was used to optimize avoiding integral windup. The PID controller follows these gains:

$$K_p = 35, K_i = 20, K_d = 20 \tag{8}$$

The Transient Mode (PD): for $|e| >= 0.1m$, this controller functions as a Type 2 system. It is maximizing the stability and speed during the initial large correction.

$$L_{pd} = (Kd \cdot s + Kp) \cdot \frac{2}{s^2} \tag{9}$$

Steady-State Mode (PID): for $|e| < 0.1m$, this controller functions as a Type 3 system. This activates the integral terms and is meant to dial in and hover accurately.

$$L_{pid} = \frac{Kd \cdot s^2 + Kp \cdot s + Ki}{s} \cdot \frac{2}{s^2} \tag{10}$$

3 Stability and Performance Analysis

Stability was verified using frequency-domain analysis (Bode and Nyquist) and verified experimentally in a PyBullet physics environment. Refer to Figures 1, 2.

3.1 Frequency Domain Results

Analysis of the open-loop transfer function $L(s) = C(s)G(s)$ yielded the following stability margins:

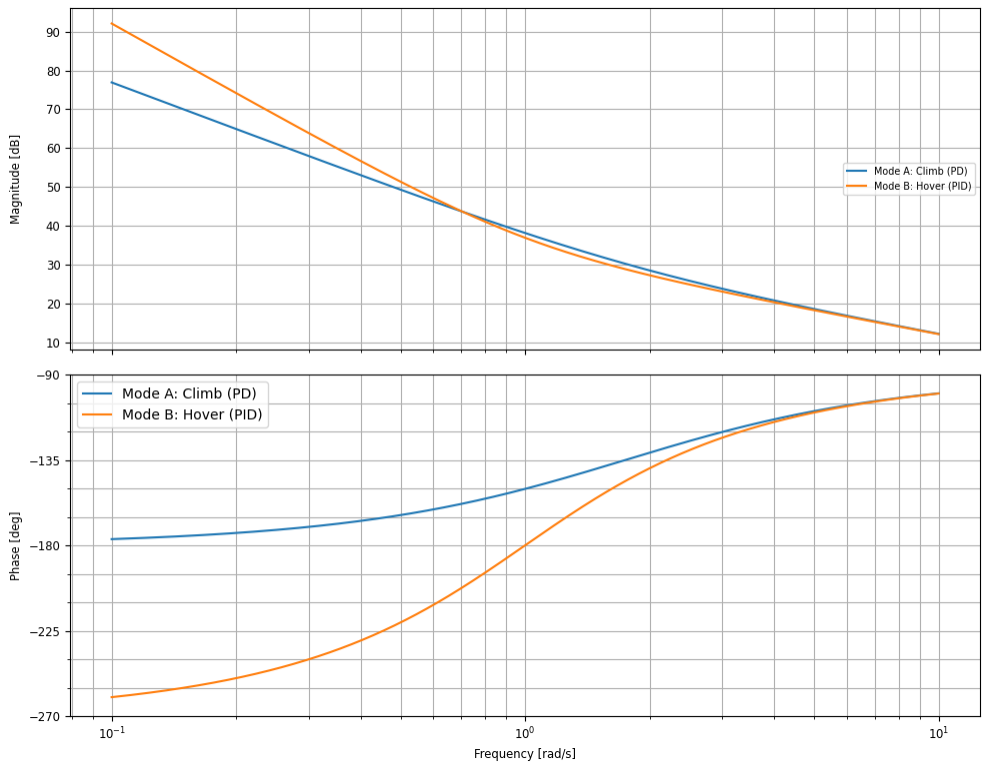

Figure 1: Phase and Gain Bode Plots

Table 1: Stability Margin Summary

Metric	Mode A (Transient)	Mode B (Steady-State)	Requirement
System Type	Type 2 (PD)	Type 3 (PID)	N/A
Gain Margin	∞ dB	-36.9 dB (Lower)	$>$ 6 dB
Phase Margin	87.5°	87.5°	$>$ 45°

3.2 Type 3 Characteristics

The transition to the second mode when the integral is added introduces an additional pole at the origin, making this altitude controller into a Type 3 system. While this provides the high-precision tracking to eliminate the steady state error it is fighting with gravity, it ends up in resulting in the complex shape seen on the Nyquist plot.

- Lead Compensation: The derivative gain ($K_d = 20$) provides sufficient phase lead to pull the response from a $-270°$ low frequency start back to stable -92.5° at the crossover frequency.
- Critical Point: The point seen on the Nyquist plot although it may seem it loops around it at (-1,0), it crosses the negative real axis at -15. This is showing zero net encirclements of the critical point, confirming the closed loop stability.

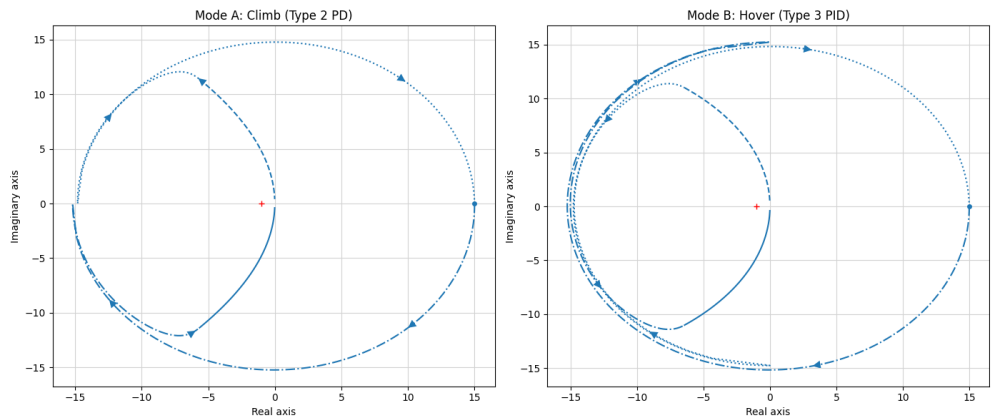

Figure 2: Nyquist Plots

- Negative Gain Margin: The -36.9 dB margin shows a much lower gain margin than usually seen of stable systems. After some further research on type 3 systems or conditionally stable systems the system will remain robust provided the gain doesn't drop below this threshold.

4 Simulation Results

The dual controller was implemented in PyBullet environment using a sampling frequency of 240Hz.

- Transient Response: The first mode really allows for the rapid ascent from 0m to 1.9m without having an integral windup that causes problems later with overshoot.
- Steady-State Accuracy: Upon entering the $|e| < 0.1m$ threshold, the second mode (PID) activates to lock the drone at the 2m rejecting gravity.
- Settling time: Although on further inspection and zooming in as shown in Figure 4, I would argue since the drone only overshoots by a max of 0.003m it has already settled once it hits 0.001m over 2m and it has settled. This performance is great and better results than was expecting.
- Code: https://github.com/abrahamAlpuerto/DroneControl

5 Conclusion

This update confirms the successful implementation of a robust SISO subsystem of the drone control. By using the conditional control logic, the system bypasses the instabilities of a Type 3 system while keeping the high accuracy steady-state benefits. Next step will be to do this for a MIMO system including pitch horizontal control!

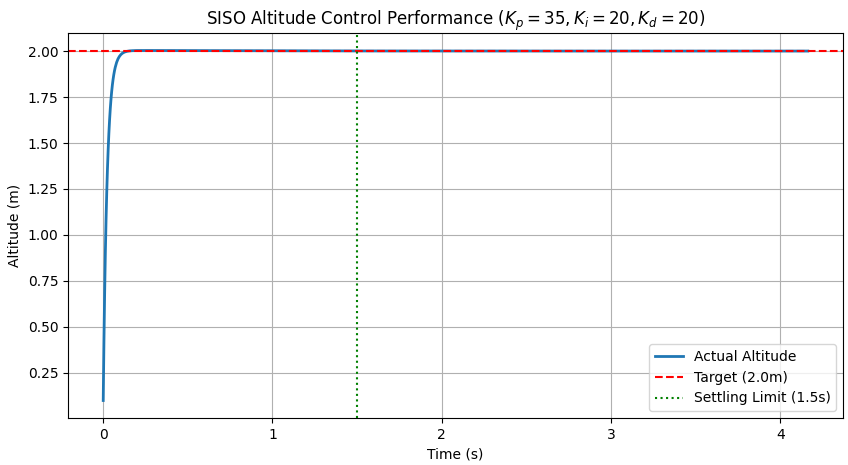

Figure 3: Drone Altitude Plot

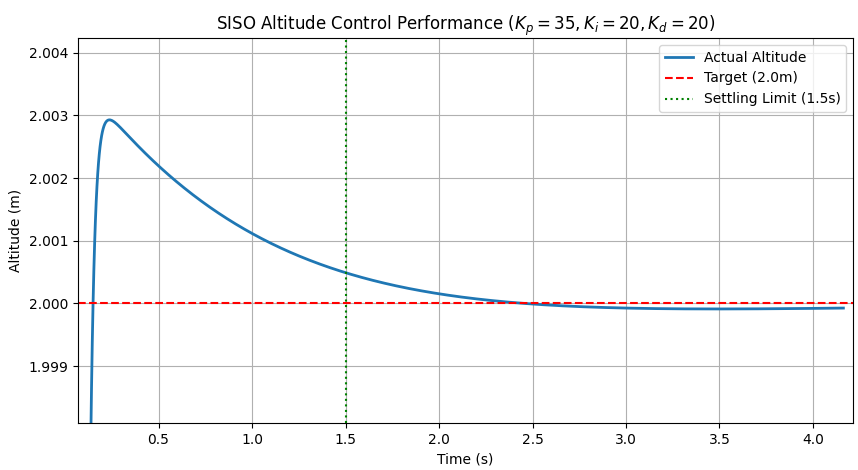

Figure 4: Drone Altitude Plot Zoomed